PROJECT REPORT

Aether:
Reference-Choice Robustness in Executable Reasoning Benchmarks

A study on Operating Systems tasks:
CPU scheduling, synchronization and Banker’s algorithm

Submitted by

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1. Dataset

- **24 tasks ("worlds")**, generated from seeded parametric specifications, in 3 families of 8:
  - CPU scheduling: FCFS, SJF, Priority, Round-Robin
  - Synchronization: mutex / semaphore interleavings
  - Banker's algorithm: safe-sequence finding
- **Valid-solution space $V(x)$:** for every task, all correct answers are enumerated exhaustively and each one is re-verified by replaying state transitions. A task is multi-valued when $|V(x)| > 1$.
- **Model outputs (main arm):** 4 local models run through Ollama, `qwen3:8b`, `gemma2:9b`, `mistral:7b-instruct`, `llama3.1:8b`.
  - 4 models $\times$ 24 worlds $\times$ 3 surface variants $\times$ 4 seeds $=$ **1,152 generations**
  - Temperature 0.7, at most 600 tokens
- **Supporting arms:**
  - 1,152 generations with the tie-breaking rule stated in the prompt (stated-convention control)
  - 576 generations from `gemma3:12b` and `olmo2:7b` (competence check)
  - 79 adversarial contrast outputs (evaluator stress test)

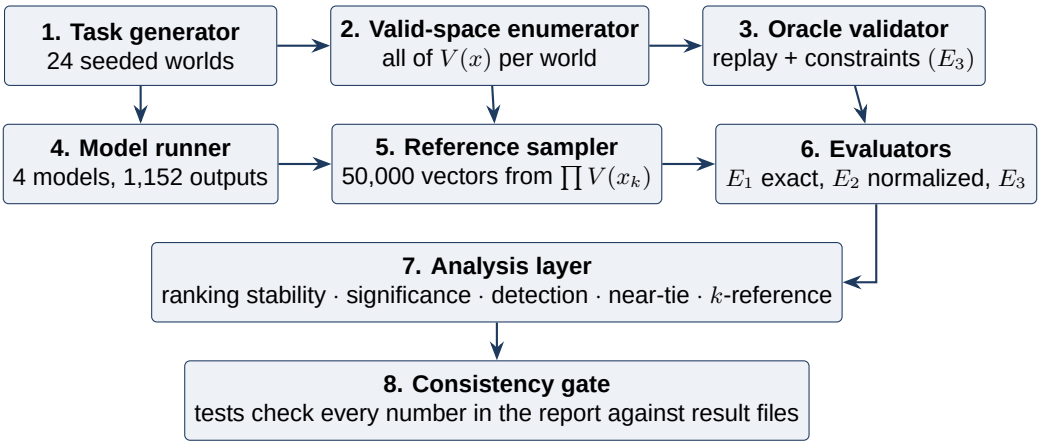
2. Dataset Preprocessing

- **Surface variants** (to avoid tying results to one wording):
  - v0: plain statement
  - v1: renamed entities, paraphrased narration
  - v2: reordered narration, table-style output
- **Statistical unit:** variants and seeds are nested inside a world, so the **world** ($N = 24$) is the unit of analysis, not the 1,152 individual generations.
- **Parsing of model text:**
  - Each response is split into steps; displayed names are mapped back to canonical entity names.
  - Unparsable output counts as a fail; truncated generations are flagged; "unverifiable" counts as not valid.
- **Normalization:**
  - Schedules are reduced to (process, start, end) segments for the normalized evaluator.
  - Whitespace is collapsed for the strict evaluator.
  - An output that mentions entities outside the task never counts as a normalized match.
- **Reference vectors:** a benchmark reference is one valid answer per world. 50,000 such vectors are drawn uniformly; "Draw 0" is the canonical (textbook tie-break) reference.

3. Code Implementation

3.1 System Architecture

No code is reproduced here; the repository holds the implementation. The system is a staged pipeline in which each stage consumes the previous stage's frozen artifact.



- **Task generator:** produces the 24 parametric worlds; seeds make them reproducible.
- **Valid-space enumerator:** lists every correct trajectory. The Banker's family uses a separate state-transition enumerator, cross-checked against brute-force permutation search on all 8 specifications.
- **Oracle validator (E_3):** replays a candidate step by step and checks the constraints. It never reads a reference, so its verdict cannot depend on which reference was chosen.
- **Model runner:** prompts each model over worlds, variants and seeds; outputs are frozen so every later analysis sees identical text.
- **Reference sampler:** draws benchmark-level reference vectors (uniform, curator-biased and similarity-biased variants).
- **Evaluators:** the three scoring rules compared in Section 3.2.
- **Analysis layer:** computes all metrics over all 50,000 draws at once, with a shared set of sign-flip permutations.
- **Consistency gate:** a 400+ test suite and a 44-claim manifest that fail the build if any reported figure drifts from the result files.

Design principles

- Freeze generations first, then vary only the reference, so any change is attributable to reference choice alone.
- Enumerate $V(x)$ exactly, so the ground-truth ranking is known and not estimated from human judgments.
- Keep the oracle independent of references and (for Banker's) independent of the solver that generated the task.

3.2 Theory Behind the Method

3.2.1 Valid-solution space

- $V(x)$ is the set of all trajectories that obey the system's transition rules and the task's constraints.
- In concurrency, independent events can be interleaved in many orders that all behave the same (equivalence classes of interleavings, Mazurkiewicz traces). Ties in scheduling and safe sequences in Banker's behave similarly.
- A benchmark that stores a single answer silently treats one member of $V(x)$ as the only truth.

3.2.2 Reference vector

- For N tasks, a benchmark's reference is one element of the product $V(x_1) \times \cdots \times V(x_N)$.
- This product is the "space of possible benchmarks" differing only in gold choice. Sampling

from it turns reference choice into a controlled experimental factor.

3.2.3 The three evaluators

- E_1 **exact match:** output must equal the reference text (after whitespace squashing). Strictest.
- E_2 **normalized match:** output must equal the reference after reducing to a canonical schedule or event trace.
- E_3 **executable validator:** output passes if it is valid, whatever the reference. Treated as ground truth, so it has zero reference dependence by construction.

3.2.4 Metrics

- **Model score:** mean pass rate over worlds.
- **Kendall τ_b:** rank agreement between the ranking under a sampled reference and under the canonical one; the b variant handles ties.
- **Winner reversal:** probability that two independent reference draws give opposite winners for a model pair.
- **E_3-ranking recovery:** share of draws where an evaluator reproduces the oracle's ordering exactly.
- **Detection:** does the evaluator find the differences the oracle finds, under a proper significance test?
- **Evaluator effect on a gap:** oracle gap minus matched-evaluator gap, with a world-level bootstrap interval.

3.2.5 Why single references lose signal (attenuation law)

- A valid output equals exactly one member of $V(x)$ and an invalid output equals none.
- A random reference therefore accepts a valid output with probability $1/|V(x)|$.
- Consequence: $\mathbb{E}[E_2] = E_3/|V(x)|$ for every task–model cell.
- The share of a model gap that survives is the oracle-gap-weighted average of $1/|V(x)|$, which is predictable from solution-space sizes before any model runs.
- The law relies on the one-match property and would fail where distinct references normalize to the same thing.

3.2.6 Significance testing

- **World-level paired sign-flip test:** per-world score differences between two models have their signs randomly flipped 20,000 times to build a null distribution. It avoids assuming normality and respects the nesting of seeds and variants.
- **Holm–Bonferroni correction** across the 6 model pairs controls false positives from multiple comparisons.
- The test is run on all 50,000 reference draws, giving a full distribution of decisions: significant for A, significant for B, or not significant.

3.2.7 Remedies analysed

- **Match-any with k references:** an output passes if it equals any of k sampled references. At $k = |V(x)|$ this equals E_3.
- **Inverse-$|V|$ correction (Horvitz–Thompson style):** rescales a single-reference score by $|V(x)|$. This removes bias but leaves very large variance, so power is not restored.
- **Executable validator:** enumerate $V(x)$ and replay candidates. Cleanest option when a validator can be written.

4. Metrics, Results and Discussion

4.1 Headline numbers

Evaluator	Kendall τ_b	Winner reversal	E_3 recovery	Mean score range
E_1 exact	0.769 ± 0.301	4.66%	2.86%	0.0–1.9%
E_2 normalized	0.618 ± 0.351	**17.56%**	13.63%	0.5–4.1%
E_3 validator	1.000	0%	100%	6.6–30.9%

- E_3 scores: `qwen3` 30.9%, `gemma2` 17.0%, `mistral` 7.6%, `llama3.1` 6.6%. All models are weak, so the whole study sits in a low-accuracy regime.
- τ_b for E_2 uses 49,913 draws; 87 draws (0.17%) have an undefined τ_b and are excluded.
- Under the single canonical reference, E_2 reproduces the oracle ordering exactly. The instability only appears when other valid references are used.

4.2 Detection of real differences

Pair	E_3 gap	E_2 gap	Gap kept	E_3 Holm p	E_1/E_2 miss it
llama3.1 vs qwen3	-0.243	-0.037	15%	0.013	100% / 99.99%
mistral vs qwen3	-0.233	-0.037	16%	0.018	100% / 99.99%
gemma2 vs qwen3	-0.139	-0.024	18%	0.200	n.s.
gemma2 vs llama3.1	$+0.104$	$+0.012$	12%	0.200	n.s.
gemma2 vs mistral	$+0.094$	$+0.012$	13%	0.200	n.s.
llama3.1 vs mistral	-0.010	0.000	-2%	0.795	n.s.

4.3 Key results

- **Reference choice alone destabilizes rankings.** E_2 flips a pairwise winner in 17.6% of reference pairs.
- **Curator bias helps but does not fix it.** Reversal falls to 12.1% (canonical chosen 50% of the time), 7.4% (80%), 2.3% (95%). Similarity-weighted sampling stays at 17.0%.
- **The larger failure is lost detection.**
  - The oracle finds `qwen3` better than `llama3.1` and `mistral` under every reference.
  - E_1 and E_2 miss both differences in 100% and 99.99% of draws, keeping only 15–16% of the gap.
  - No wrong-direction significant result occurs: the loss is power, not sign.
- **Near-ties dominate reversals.** 85.6% of E_2 reversal mass is in pairs the oracle cannot separate (22.1% reversal there vs 7.4% in separated pairs).
- **By family:** Banker's 21.0%, synchronization 10.0%, scheduling 0% (uninformative, since three of four models solve none).
- **Stating the tie-break rule does not help.** 62.5% of semantically valid outputs are still rejected; `gemma3:12b` gives 61.9% ($n_{\text{valid}} = 42$).
- **Adversarial set:** E_1/E_2 reject 100% of valid-but-noncanonical answers; E_3 accepts them (this is true by definition of E_3, so it is a stress test and not an error-rate estimate).

4.4 Remedy results (match-any with k references)

k	1	5	10	25	all (E_3)
Detect qwen3 $>$ llama3.1	0%	48%	58%	84%	100%
Detect qwen3 $>$ mistral	0%	57%	79%	88%	100%

- False positives on non-separated pairs stay near or below the nominal 5%.
- The inverse-$|V|$ correction reaches only 40% / 31% at $k = 10$, below match-any (58% / 79%).

4.5 Discussion

- Single-gold matching in multi-solution domains mostly **hides real differences** and **makes weak orderings arbitrary**; it rarely reorders strong ones.
- The attenuation law explains why: each valid answer survives only $1/|V(x)|$ of the time.
- The executable validator removes the dependence by construction; the study's value is measuring the cost of not using one.
- Scope: one 24-world benchmark, four sub-10B models, so findings are a case study and not a general claim.

5. What Went Wrong

- **Wrong significance claim.**
  - An early cell-level McNemar test treated seeds and variants as independent (pseudoreplication) and reported "50% significance flips". Replaced by the world-level test.
  - A later summary, "no pair is significant", was true only for E_1/E_2; E_3 separates two pairs. Fixed by rerunning all 50,000 draws per evaluator.
- **Bad citations.** Three arXiv IDs pointed to unrelated papers and one title/author list was wrong. All references were re-verified and a test now blocks mismatches. "First" and "genuinely new" claims were removed because ranking instability under reference changes is already published.
- **Benchmark leak.** The first trace bank let a distance-to-reference classifier separate valid from invalid traces (0.663 vs a 0.65 limit); it was redesigned before any model ran. Two parser bugs were fixed after reading real outputs.
- **Scoring-rule bug.** The analysis code omitted the rule that outputs naming outside entities never match, so 7 `mistral` outputs were wrongly counted. Fixing it moved τ_b from 0.490 to 0.618, reversal from 18.61% to 17.56% and recovery from 23.92% to 13.63%. All results were regenerated and the change is disclosed.
- **Weak models, small benchmark.**
  - Models solve only 7–31% of tasks, so most pairs cannot be separated.
  - The pilot's own pre-specified verdict was CONDITIONAL, and its follow-up conditions (models $\geq$50% valid, $\geq$65 worlds) were not met.
  - The RCR metrics are a post-hoc reanalysis, and the "preregistration" was an internal freeze, not an external registration.
- **Smaller slips:** a 200-draw significance subsample (superseded) and 87 draws with undefined τ_b that must be excluded.

6. Alternative Flow (Improvements)

1. **Done: explain the effect.** Attenuation law $\mathbb{E}[E_2] = E_3/|V(x)|$ matched observations with $r = 1.000$ over 42 task–model cells.
2. **Done: k-reference evaluation.** Detection reaches 84–88% at $k = 25$.
3. **Done: bias correction test.** Inverse-$|V|$ scaling is unbiased but too noisy.
4. **Next: more statistical power.** At least 65 worlds and 10+ models, including stronger ones, so more pairs become separable.
5. **Next: second domain.** Topological orderings of dependency graphs have exactly countable $|V|$ and would test generality.
6. **Next: LLM judges.** Check whether reference-guided judges show the same sensitivity to reference swaps.
7. **Next: external registration.** Publish the follow-up plan with an independent timestamp before running it.